CP-AnemiC version 1

710 rows: 424 anaemic (60 %), 286 non-anaemic

Stage 1: SHA-256 file digests

498 distinct files
303 rows in 91 groups of byte-identical files

Metadata inside the groups

Hb differs in 90 of the 91 byte-identical groups;
the largest holds 10 copies with 9 Hb values

Stage 2: perceptual hash (image and mirror)

256-bit pHash, Hamming distance of at most 20 bits
26 further files linked, in 13 groups

Largest connected component

11 rows, 11 Hb values, 9 facilities
(10 byte-identical copies + 1 mirrored copy)

472 unique images (pre-specified map)

386 singleton rows + 86 multi-row groups (324 rows)
1 group with conflicting labels excluded
Classification set: 471 images (229 anaemic, 242 not)
Singleton set for Hb regression: 386 images

Class dependence

Duplicated rows 81 % anaemic, singletons 42 %
85 of 86 groups carry one label;
permutation null 25.4 (maximum 41), $p < 0.0001$

Stage 3 (post hoc): pHash under 8 rotations and reflections + identical colour features

19 pairs of groups merged (rotated copies)
453 unique images; strict set: 452 (218 anaemic, 234 not)

Rotated copies

All 19 pairs agree on the anaemia label
but not on Hb

Stage 4 (post hoc): SIFT keypoints + RANSAC on all 111,156 pairs, photometric verification

168 new pairs of images of the same eye
(re-cropped, rescaled or mirrored)
338 distinct images (204 singletons + 134 groups)
Geometric set: 337 images (179 anaemic, 158 not)

Images of the same eye

All 168 pairs agree on the anaemia label
(permutation null 84, $p < 0.0001$);
142 differ in facility and 144 in Hb